The documentation and process conversion measures necessary to comply with this revision shall be completed by 19 November 1994

INCH-POUND

MIL-S-19500/446A
19 August 1994
SUPERSEDING
MIL-S-19500/446
9 February 1971

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, SILICON, HIGH-POWER, SINGLE PHASE, FULL WAVE BRIDGE RECTIFIER, JAN, JANTX, AND JANTXV TYPE SPA25, SPB25, SPC25, SPD25

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, single phase full wave rectifiers, intended for use in applications at frequencies of 1 KHz or less. Three levels of product assurance are provided for each device type as specified in MIL-S-19500 (See 6.3).

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

Type number	V_{RWM}	Bridge 1/ I_{O1} at T_C = 55°C	Bridge 2/ I_{O2} at T_C = 100°C	$I_{F(surge)}$ I_O = 25 A dc T_C = 55°C t_p = 8.3 ms	Barometric pressure reduced	T_{rr} I_F = 0.5 A I_R = 1.0A I_{rec} = 0.25 A
	V (pk)	A dc	A dc	A (pk)	mmHg	µs
SPA25	100	25	15	150	8	2.5
SPB25	200	25	15	150	8	2.5
SPC25	400	25	15	150	8	2.5
SPD25	600	25	15	150	8	2.5

1/ Derate from 25 A dc at 55°C to 15 A dc at 100°C (222 mA dc/°C).

2/ Derate from 15 A dc at 100°C to 0 A dc at 150°C (300 mA dc/°C).

Operating temperature: -65°C to +150°C
Storage ambient temperature -65°C to +150°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5961

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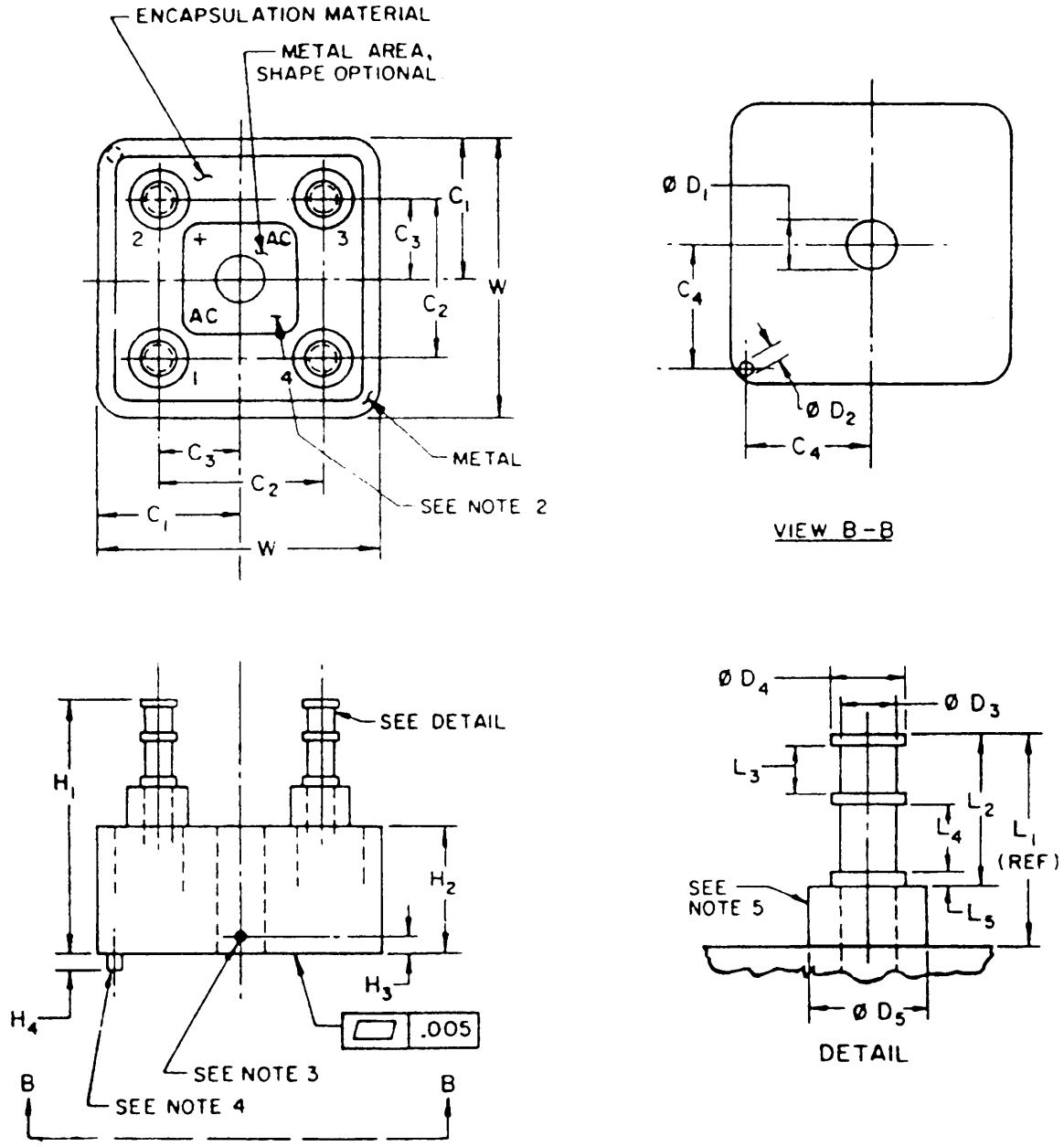


FIGURE 1. Physical dimensions.

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Dimensions					
Ltr.	Inches		Millimeters		Notes
	Min.	Max.	Min.	Max.	
C_1	.552	.572	14.02	14.53	
C_2	.624	.760	15.85	19.30	
C_3	.312	.380	7.92	9.65	
C_4	.495	.512	12.57	13.00	
ϕD_1	.189	.195	4.80	4.95	
ϕD_2	.057	.067	1.45	1.70	
ϕD_3	.108	.118	2.74	3.00	
ϕD_4	.141	.151	3.58	3.84	
ϕD_5	.225	.260	5.72	6.60	
H_1	.690	1.060	17.53	26.92	
H_2	.300	.500	7.62	12.70	
H_3	.040	.060	1.02	1.52	
H_4	.042	.062	1.07	1.57	
L_1	.370	.560	9.40	14.22	
L_2	.307	.365	7.80	9.27	
L_3	.089	.099	1.49	2.49	
L_4	.132	.163	3.35	4.14	
L_5	.026	.036	.66	.91	
W	1.104	1.144	28.04	29.06	

NOTES:
1. Dimensions are in inches. Metric equivalents are given for general information only.
2. Polarity shall be marked on terminal side of the device. Terminal numbers are for reference and do not have to be marked on the bridge;
3. Point at which T_C is read shall be in metal part of case as shown on drawing.
4. The locating pin shall be adjacent to the positive (+) terminal.
5. Insulating sleeve shall be alumina (AL_2O_3) composite material or equivalent.
6. The areas defined by the dimensions "L3" and "L4" are both acceptable for wire wrap and soldering.

FIGURE 1. Physical dimensions - continued.

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applied in more than one layer. The adjacent layers

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1.

3.3.1 Internal construction. The rectifier bridge shall consist of a metal and plastic encased assembly of discrete diodes. Each discrete diode shall be a glass-to-metal, ceramic-to-metal, or fused metal oxide-to-metal hermetically sealed package. No multiple diodes per leg construction shall be permitted. The silicon die in each discrete diode shall be metallurgically bonded directly to the terminal pins. The completed assembly of diodes and other internal structures shall be encapsulated in a plastic material which polymerizes to a rigid condition by virtue of a chemical cross-linking mechanism. The rectifier bridge shall be free of voids either visible or as evidenced by failure to pass the environment test specified herein. Only those discrete diodes which have met these requirements shall be used in the rectifier bridge. Discrete diodes shall be manufactured and tested by the rectifier bridge manufacturer.

3.3.2 Terminal finish. Terminal finish shall be in accordance with MIL-S-19500.

3.4 Marking. Devices shall be marked as specified in MIL-S-19500.

3.4.1 Polarity. Polarity shall be as marked in figure 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JAN, JANTX, and JANTXV). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable. The JANTXV requirements apply to the internal discrete diodes only, not to the assembly.

4.3.1 Discrete Diode Screening. 100 percent of the internal discrete diodes shall be subjected to the following.

Screen (see table II of MIL-S-19500)	Measurement
	JAN, JANTX, and JANTXV
1/	Thermal Impedance (see 4.5.3)
9	Not applicable
11	I_{R1} and V_{F1}
12	See 4.3.1.1
13 2/	Subgroup 2 of table I herein. 2/ ΔV_{F1} = ±0.10 V dc; I_{R1} = 100 percent of initial value or ± 250 nA dc, whichever is greater.

1/ Shall be performed anywhere after screen 3.
2/ Except thermal impedance, if already performed

4.3.1.1 Power burn in conditions. Power burn in conditions are as follows: MIL-STD-750 method 1038, test condition B, TA = room ambient as defined in the general requirements of paragraph 4.5 of MIL-STD-750, V_{RWM} = rated V_{RWM} (see 1.3), I_O = 3 A dc, f = 50 - 60 Hz.

4.3.2 Assembly screening. 100 percent of the completed assemblies shall be subjected to the following.

Screen	Measurement
	JAN, JANTX, and JANTXV
	MIL STD 750, method 1051, condition F
End points 3/	Subgroup 2 of table I herein
Dielectric withstanding voltage	See 4.5.1

3/ Except thermal impedance.

4.4 Quality Conformance inspection. Quality conformance inspection shall be performed on the finished rectifier assemblies and in accordance with MIL-S-19500.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table I, subgroup 2 (except thermal impedance) herein.

4.4.2.1 Group B inspection, table IVb of MIL-S-19500

Subgroup	Method	Conditions
B1	2026	Immerse to insulating sleeve
B2	1051	Condition F, 25 cycles.
B2	1071	Not applicable
B3	1027	I_O = 0 A dc; T_A = 150°C; V_{RWM} = 80% rated V_{RWM} (see 1.3), f = 60Hz Sampling plan; LTPD = 10
B5		Operational thermal cycling, (see 4.5.5).
B5	3105	Junction Temperature Test (see 4.5.2)
B6		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table I, subgroup 2 (except thermal impedance) herein.

Subgroup	Method	Conditions
C2	1056	Not applicable
C2	2036	Test condition A, 6 pounds, t = 30 seconds.
C2	2036	Test condition D1, t = 10 seconds, 5 inch-pounds.
C2	1071	Not applicable.
C5	1001	Pressure = 8 mm Hg; I_R = 2.0uA dc maximum, V_R = rated V_{RWM}(see 1.3),(applied between all terminals to the case), t = 60 seconds; sampling plan: LTPD = 15, c = 0.
C6	1026	I_O = 0 A dc; T_A = 150°C ; V_{RWM} = 80% rated V_{RWM} (see 1.3), f = 60 Hz.
C7	4066	V_{RWM} = 0 V, bridge I_O = 25 A dc (entire bridge biased); T_C = 55°C + 10°C,- 0°C; I_{FSM} = 150 A (pk) (each device per bridge leg); t_p = 8.3 ms; 10 surges per leg at maximum 1 minute intervals; sampling plan: LTPD = 10. Alternate condition: Each leg may be biased separately to Io = 12.5 Adc per leg. All other conditions the same as above.

4.5 Methods of Inspection. Methods of inspection shall be as specified in the appropriate tables as follows:

4.5.1 Dielectric withstanding voltage. This test shall be performed with the metal case of the assembly connected to ground and all terminals connected to the high potential side of a DC power supply or a scope display test set. The voltage applied between the terminals and the case, shall be 2,800 volts DC. Any discontinuity or dynamic instability of the trace, or a breakdown current in excess of 10 μAdc shall be cause rejection.

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4.5.2 Junction temperature test. This test shall be performed in accordance with method 3105 of MIL-STD-750. The maximum junction temperature for any diode in the bridge at the below specified conditions shall be 175°C maximum. The test conditions shall be as follows:

T_C = 55°C	or T_C = 100°C
I_O = 25 A dc	I_O = 15 Adc
I_{ref} = 10 mA dc	I_{ref} = 10 mA dc

4.5.3 Thermal response. $Z_{\theta JX}$ measurement shall be performed in accordance with method 3101 of MIL-STD-750. The following parameters shall apply:

a. Measurement current (I_M) . . . . . 10 mA.

b. Heating current (I_H) . . . . . 10 A minimum.

c. Heating time (t_H) . . . . . 10 ms.

d. Measurement time delay (t_{MD}) . . . . . 100 us maximum.

The maximum limit for $Z_{\theta JX}$ under these test conditions is 1.5 °C/watt.

4.5.4 Operational thermal cycling. This test shall be performed by connecting a single phase 50-60 Hz sine wave input to the ac terminals of the bridge. The resistive load or input voltage shall be made variable in order to maintain a rated bridge output current of 25 amperes DC. Bridge case temperature shall be maintained at 55°C except during the cooling cycle when case temperature shall be permitted to drop 20°C maximum. Life-test duration shall be 2,000 cycles with a heating time of 8 minutes, +10, -0 minutes and a cooling time $\geq$ 2 minutes.

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TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal Impedance /leg	3101	see 4.5.3	$Z_{\theta JX}$		1.5	°C/W
Forward voltage /leg	4011	I_F = 39 A dc (pk) $t_p \leq$ 8.3 ms; duty cycle $\leq$ 2 percent	V_{F1}	0.9	1.40	V (pk)
Reverse current /leg	4016	DC method; V_R = rated V_{RWM} (see 1.3)	I_{R1}		2.0	µA dc
Breakdown voltage /leg	4021	I_R = 50 µ A dc	V(BR)1			
SPA25				110		Vdc
SPB25				220		Vdc
SPC25				440		Vdc
SPD25				660		Vdc
Subgroup 3						
High temperature operation:		T_C = 100°C				
Reverse current /leg	4016	DC method; V_R = rated V_{RWM} (see 1.3)	I_{R2}		150	µA dc
Low temperature operation:		T_A = -55°C				
Reverse current /leg	4016	DC method; V_R = rated V_{RWM} (see 1.3)	I_{R3}		50	µA dc
Subgroup 4						
Reverse Recovery Time /leg	4061	Condition B I_F = 0.5 A; I_R = 1.0A I_{rec}= 0.25A	T_{rr}		2.5	uS
Subgroup 5,6		Not applicable				

1/ For sampling plan, see MIL-S-19500.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Terminal finish as specified (see 3.3.2).

c. Product assurance level and type designation.

6.3 Interchangeability. JAN devices previously manufactured to MIL-S-19500/446 were constructed with JANTX level devices and are the same product assurance level as the JAN and JANTX devices in this revision. JAN and JANTX devices are identical for all requirements of this specification except the device marking and are directly interchangeable. The JANTX device is the preferred part number for new design and procurement.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-1639)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/446A	2. DOCUMENT DATE (YYMMDD) 94/08/19

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, SILICON, HIGH-POWER, SINGLE PHASE, FULL WAVE BRIDGE RECTIFIER, JAN, JANTX, and JANTXV TYPE SPA25, SPB25, SPC25, SPD25

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048	(2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center ATTN: DESC-ELDT Dayton, OH 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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